

Tables

A

Table A.1 Conversion Factors

Length

	m	cm	km	in.	ft	mi
1 meter	1	10^2	10^{-3}	39.37	3.281	6.214×10^{-4}
1 centimeter	10^{-2}	1	10^{-5}	0.393 7	3.281×10^{-2}	6.214×10^{-6}
1 kilometer	10^3	10^5	1	3.937×10^4	3.281×10^3	0.621 4
1 inch	2.540×10^{-2}	2.540	2.540×10^{-5}	1	8.333×10^{-2}	1.578×10^{-5}
1 foot	0.304 8	30.48	3.048×10^{-4}	12	1	1.894×10^{-4}
1 mile	1 609	1.609×10^5	1.609	6.336×10^4	5 280	1

Mass

	kg	g	slug	u
1 kilogram	1	10^3	6.852×10^{-2}	6.024×10^{26}
1 gram	10^{-3}	1	6.852×10^{-5}	6.024×10^{23}
1 slug	14.59	1.459×10^4	1	8.789×10^{27}
1 atomic mass unit	1.660×10^{-27}	1.660×10^{-24}	1.137×10^{-28}	1

Note: 1 metric ton = 1 000 kg.

Time

	s	min	h	day	yr
1 second	1	1.667×10^{-2}	2.778×10^{-4}	1.157×10^{-5}	3.169×10^{-8}
1 minute	60	1	1.667×10^{-2}	6.994×10^{-4}	1.901×10^{-6}
1 hour	3 600	60	1	4.167×10^{-2}	1.141×10^{-4}
1 day	8.640×10^4	1 440	24	1	2.738×10^{-5}
1 year	3.156×10^7	5.259×10^5	8.766×10^3	365.2	1

Speed

	m/s	cm/s	ft/s	mi/h
1 meter per second	1	10^2	3.281	2.237
1 centimeter per second	10^{-2}	1	3.281×10^{-2}	2.237×10^{-2}
1 foot per second	0.304 8	30.48	1	0.681 8
1 mile per hour	0.447 0	44.70	1.467	1

Note: 1 mi/min = 60 mi/h = 88 ft/s.

Force

	N	lb
1 newton	1	0.224 8
1 pound	4.448	1

(Continued)

Table A.1 Conversion Factors (*continued*)**Energy, Energy Transfer**

	J	ft · lb	eV
1 joule	1	0.737 6	6.242×10^{18}
1 foot-pound	1.356	1	8.464×10^{18}
1 electron volt	1.602×10^{-19}	1.182×10^{-19}	1
1 calorie	4.186	3.087	2.613×10^{19}
1 British thermal unit	1.055×10^3	7.779×10^2	6.585×10^{21}
1 kilowatt-hour	3.600×10^6	2.655×10^6	2.247×10^{25}

	cal	Btu	kWh
1 joule	0.238 9	9.481×10^{-4}	2.778×10^{-7}
1 foot-pound	0.323 9	1.285×10^{-3}	3.766×10^{-7}
1 electron volt	3.827×10^{-20}	1.519×10^{-22}	4.450×10^{-26}
1 calorie	1	3.968×10^{-3}	1.163×10^{-6}
1 British thermal unit	2.520×10^2	1	2.930×10^{-4}
1 kilowatt-hour	8.601×10^5	3.413×10^2	1

Pressure

	Pa	atm
1 pascal	1	9.869×10^{-6}
1 atmosphere	1.013×10^5	1
1 centimeter mercury ^a	1.333×10^3	1.316×10^{-2}
1 pound per square inch	6.895×10^3	6.805×10^{-2}
1 pound per square foot	47.88	4.725×10^{-4}

	cm Hg	lb/in.²	lb/ft²
1 pascal	7.501×10^{-4}	1.450×10^{-4}	2.089×10^{-2}
1 atmosphere	76	14.70	2.116×10^3
1 centimeter mercury ^a	1	0.194 3	27.85
1 pound per square inch	5.171	1	144
1 pound per square foot	3.591×10^{-2}	6.944×10^{-3}	1

^aAt 0°C and at a location where the free-fall acceleration has its “standard” value, 9.806 65 m/s².**Table A.2** Symbols, Dimensions, and Units of Physical Quantities

Quantity	Common Symbol	Unit^a	Dimensions^b	Unit in Terms of Base SI Units
Acceleration	$\vec{a}$	m/s ²	L/T ²	m/s ²
Amount of substance	n	MOLE		mol
Angle	θ, ϕ	radian (rad)	1	
Angular acceleration	$\vec{\alpha}$	rad/s ²	T ⁻²	s ⁻²
Angular frequency	ω	rad/s	T ⁻¹	s ⁻¹
Angular momentum	$\vec{L}$	kg · m ² /s	ML ² /T	kg · m ² /s
Angular velocity	$\vec{\omega}$	rad/s	T ⁻¹	s ⁻¹
Area	A	m ²	L ²	m ²
Atomic number	Z			
Capacitance	C	farad (F)	Q ² T ² /ML ²	A ² · s ⁴ /kg · m ²
Charge	q, Q, e	coulomb (C)	Q	A · s

(Continued)

Table A.2 Symbols, Dimensions, and Units of Physical Quantities (*continued*)

Quantity	Common Symbol	Unit ^a	Dimensions ^b	Unit in Terms of Base SI Units
Charge density				
Line	λ	C/m	Q/L	A · s/m
Surface	σ	C/m ²	Q/L ²	A · s/m ²
Volume	ρ	C/m ³	Q/L ³	A · s/m ³
Conductivity	σ	1/Ω · m	Q ² T/ML ³	A ² · s ³ /kg · m ³
Current	I	AMPERE	Q/T	A
Current density	J	A/m ²	Q/TL ²	A/m ²
Density	ρ	kg/m ³	M/L ³	kg/m ³
Dielectric constant	κ			
Electric dipole moment	$\vec{p}$	C · m	QL	A · s · m
Electric field	$\vec{E}$	V/m	ML/QT ²	kg · m/A · s ³
Electric flux	Φ_E	V · m	ML ³ /QT ²	kg · m ³ /A · s ³
Electromotive force	$\mathcal{E}$	volt (V)	ML ² /QT ²	kg · m ² /A · s ³
Energy	E, U, K	joule (J)	ML ² /T ²	kg · m ² /s ²
Entropy	S	J/K	ML ² /T ² K	kg · m ² /s ² · K
Force	$\vec{F}$	newton (N)	ML/T ²	kg · m/s ²
Frequency	f	hertz (Hz)	T ⁻¹	s ⁻¹
Heat	Q	joule (J)	ML ² /T ²	kg · m ² /s ²
Inductance	L	henry (H)	ML ² /Q ²	kg · m ² /A ² · s ²
Length	ℓ, L	METER	L	m
Displacement	$\Delta x, \Delta \vec{r}$			
Distance	d, h			
Position	$x, y, z, \vec{r}$			
Magnetic dipole moment	$\vec{\mu}$	N · m/T	QL ² /T	A · m ²
Magnetic field	$\vec{B}$	tesla (T) (= Wb/m ²)	M/QT	kg/A · s ²
Magnetic flux	Φ_B	weber (Wb)	ML ² /QT	kg · m ² /A · s ²
Mass	m, M	KILOGRAM	M	kg
Molar specific heat	C	J/mol · K		kg · m ² /s ² · mol · K
Moment of inertia	I	kg · m ²	ML ²	kg · m ²
Momentum	$\vec{p}$	kg · m/s	ML/T	kg · m/s
Period	T	s	T	s
Permeability of free space	μ_0	N/A ² (= H/m)	ML/Q ²	kg · m/A ² · s ²
Permittivity of free space	ϵ_0	C ² /N · m ² (= F/m)	Q ² T ² /ML ³	A ² · s ⁴ /kg · m ³
Potential	V	volt (V)(= J/C)	ML ² /QT ²	kg · m ² /A · s ³
Power	P	watt (W)(= J/s)	ML ² /T ³	kg · m ² /s ³
Pressure	P	pascal (Pa)(= N/m ²)	M/LT ²	kg/m · s ²
Resistance	R	ohm (Ω)(= V/A)	ML ² /Q ² T	kg · m ² /A ² · s ³
Specific heat	c	J/kg · K	L ² /T ² K	m ² /s ² · K
Speed	v	m/s	L/T	m/s
Temperature	T	KELVIN	K	K
Time	t	SECOND	T	s
Torque	$\vec{\tau}$	N · m	ML ² /T ²	kg · m ² /s ²
Velocity	$\vec{v}$	m/s	L/T	m/s
Volume	V	m ³	L ³	m ³
Wavelength	λ	m	L	m
Work	W	joule (J)(= N · m)	ML ² /T ²	kg · m ² /s ²

^aThe base SI units are given in uppercase letters.^bThe symbols M, L, T, K, and Q denote mass, length, time, temperature, and charge, respectively.